Home Lab 3: The Starch and Carbonate Tests

Do this with a parent. Goggles on. Never taste anything. Clean up after.

Goal

See exactly what the iodine test and the acid-fizz test do — including the tricky results (powdered sugar, baking powder, vitamin C).

You need

From the kit: **corn starch, granulated sugar, powdered sugar, calcium carbonate (chalk), baking soda**; iodine. From home: baking **powder**, a crushed vitamin C tablet (or a pinch of the vitamin C from the kit if included) dissolved in a little water, white vinegar, water, toothpicks, cups, paper towels.

Part 1: Iodine

Put a dry pinch of each on a paper towel. Add **one drop of iodine** to each. **Predict first**, then test.

Powder	My prediction	Actual (blue-black / orange / colorless)
Corn starch		
Granulated sugar		
Powdered sugar		
Baking powder		
Baking soda		
Vitamin C solution (drop iodine into it)		

Part 2: Acid fizz

Put a fresh pinch of each on a paper towel and add a few drops of vinegar.

Powder	Fizz?	How strong?
Baking soda		
Calcium carbonate (chalk)		
Corn starch		
Baking powder (add water first, then vinegar)		

Follow-up

1. Powdered sugar gave a faint blue with iodine even though sugar has no starch. Why?
2. Baking powder turned blue-black but baking soda didn't. What does baking powder contain that baking soda doesn't?
3. What happened when you added iodine to the vitamin C solution? Is that the same kind of thing as the starch test?
4. Chalk and baking soda both fizzed. Which fizzed faster, and why?

Coach notes

- Iodine: corn starch, powdered sugar (faint — has ~3% corn starch), baking powder → blue-black. Granulated sugar, baking soda → stays orange. Vitamin C → **iodine goes colorless** (vitamin C is a reducing agent; a real reaction, unlike the starch inclusion complex).
- Fizz: baking soda strong and fast; chalk fizzes but slower (insoluble — only the grain surface reacts); baking powder fizzes with water alone (contains a dry acid) and more with vinegar; corn starch doesn't fizz.
- Follow-up: (1) corn starch is added to powdered sugar as an anti-caking agent. (2) corn starch. (3) the iodine is chemically reduced to colorless iodide — a reaction, not an inclusion complex. (4) baking soda — it dissolves so all of it reacts at once; chalk is insoluble.